

# Rapid System Prototyping and Validation with the RealView® SoC Designer Tool

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ESL Product Manager



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## Agenda

- Introduction to ESL Design
- Gating Factors for ESL Design Success
- Introduction of the RealView SoC Designer Tool
- System Visibility and Validation
- Reuse and Standardization
- Conclusion

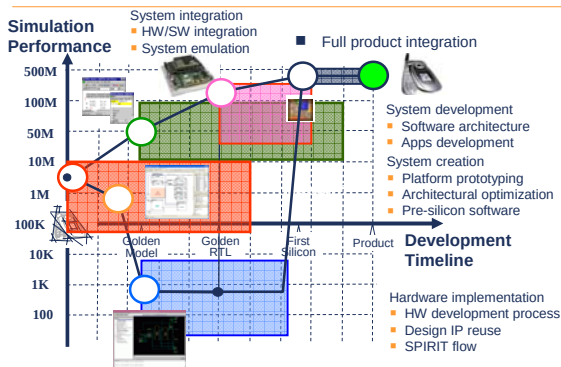


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## The Product Life-Cycle

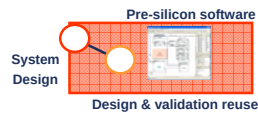


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## ESL Tackles the Design Challenge



- Validate system architectures against product requirements prior to RTL implementation
- Reduce cost by avoiding over-designing
- Eliminate silicon re-spins by avoiding under-design
- Parallelize hardware and software design tasks, accelerating time to market
- Exploit the value of designing with third-party IP
- Bridge between product requirements capture and SoC implementation



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## Gating Factors for ESL Design Success

- Prototypes must be created with reasonable effort
  - Availability of ESL representations of third-party IP
  - Reasonable effort for custom component modeling
  - Reuse of component models, platform prototypes and validation IP
- Must have an environment for rapid platform prototyping, analysis and debug
  - Integrated with software tool chains
  - System debug, visualization and validation mechanisms
  - Deployable for architectural, validation and software groups
- Prototypes must be accurate
  - Accuracy of third-party IP must be assured



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## The RealView SoC Designer Tool

- Fast professional design entry and simulation tool accelerates system design and validation!
- High-performance cycle-based SystemC simulator
- Synchronous multi-core simulation for system-level virtual prototyping
- Extensive hardware and software visibility into your SoC design
- Enables software development for early hardware and software integration
- Robust ecosystem of partner tools to support your design flow
- Incorporates MaxSim™ technology



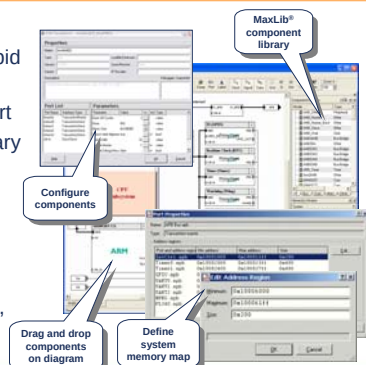
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## The SoC Designer Canvas

- Easy-to-use block diagram editor for rapid SoC prototyping
- Full multi-core support
- Extensive model library of components
- SystemC and C++ Component Wizard
- SystemC component import and export
- Support for Windows, Solaris and Linux



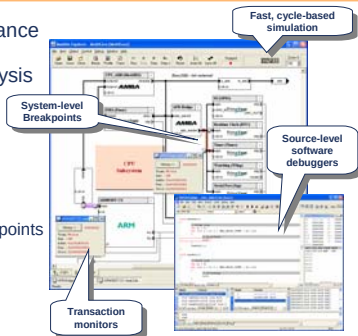
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## The SoC Designer Simulator

- Validate SoC performance against spec
- Perform trade-off analysis of different
  - Cores
  - Bus architectures
  - Caches
  - Memories
- SoC debugging
  - Breakpoints and watchpoints
  - Runtime visualizations
  - Simulation control
- Software debugging
  - Fast execution
  - Disassembly and source level debug




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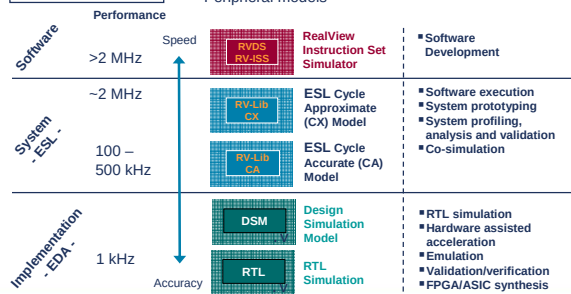
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## RealView Model Library - SoC Designer

- Processors (ARM, Ceva, LSI Logic, StarCore, etc...)
- Bus and memory subsystem
- Peripheral models




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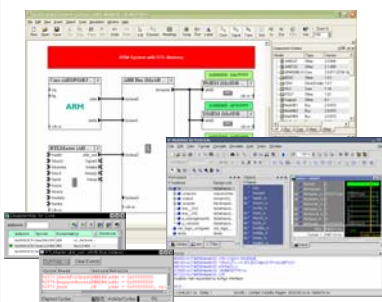
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## Co-Design Package

- Reuse system models and system level verification IP



System Level Design and Verification



- HDL Co-Sim
- Transactors
- VIP Reuse



Synthesis Flow

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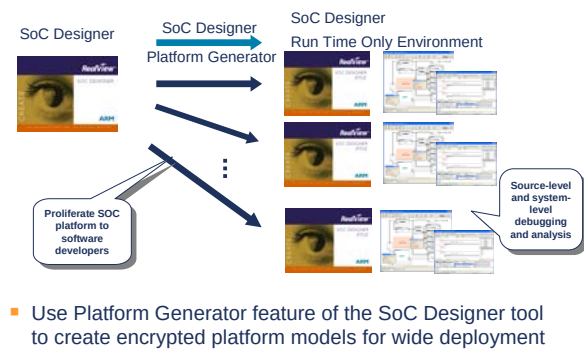
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## Run-Time Only Environment

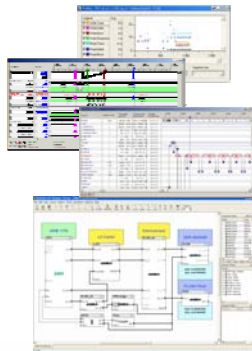


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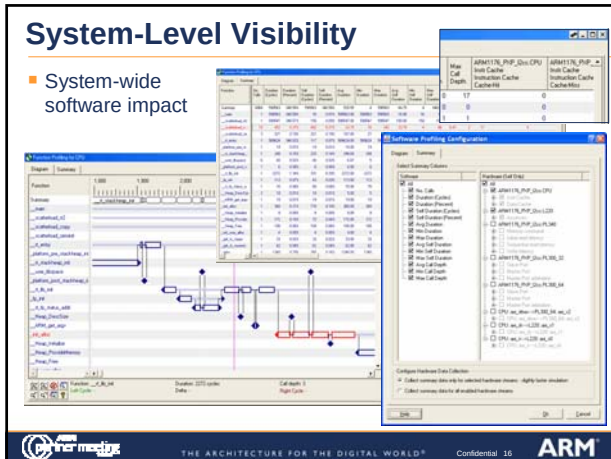
## System Visibility and Validation

- Rising failure rates attributed to lack of system-level visibility
- What is needed:
  - End-to-end system performance
  - Software execution in system context
  - Realistic algorithm execution
  - Directed and random traffic generation
  - System-level regression testing



## System-Level Visibility

- System-wide software impact




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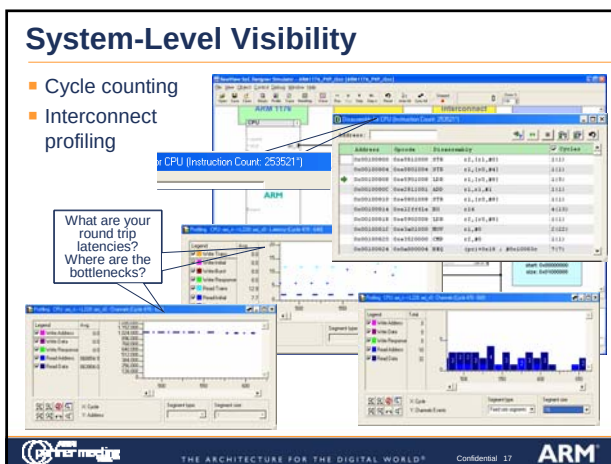
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## System-Level Visibility

- Cycle counting
- Interconnect profiling




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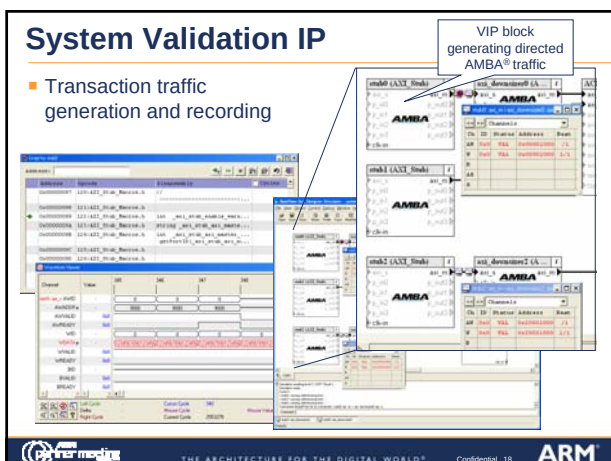
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## System Validation IP

- Transaction traffic generation and recording




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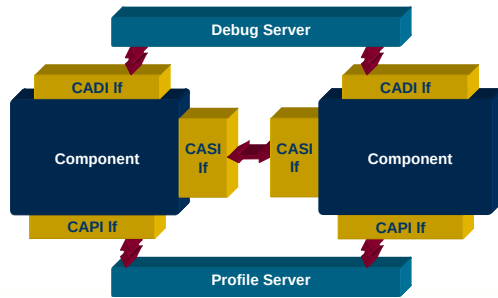
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## ESL Interface Standards

Standard simulation API key to rapid prototyping and IP reuse

- RealView ESL API Overview

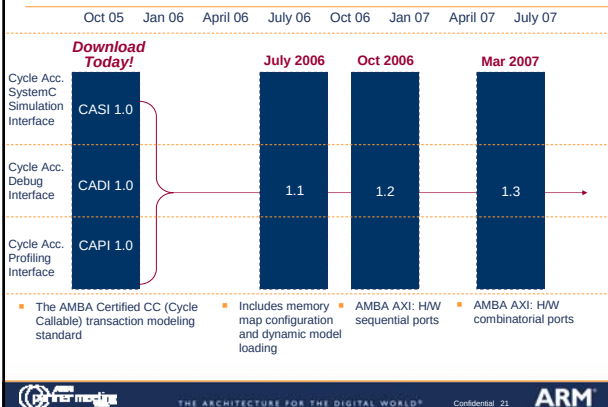


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## Interface Technology Timeline

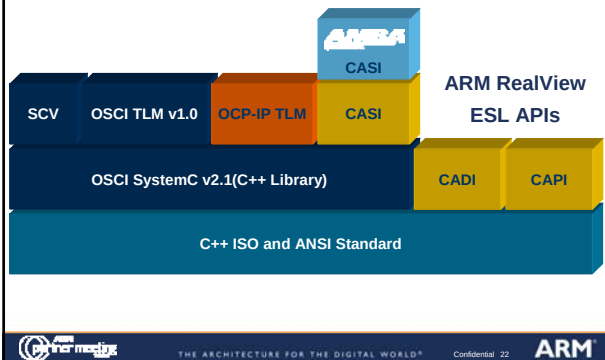


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## I/F Industry Standardization: SystemC Today




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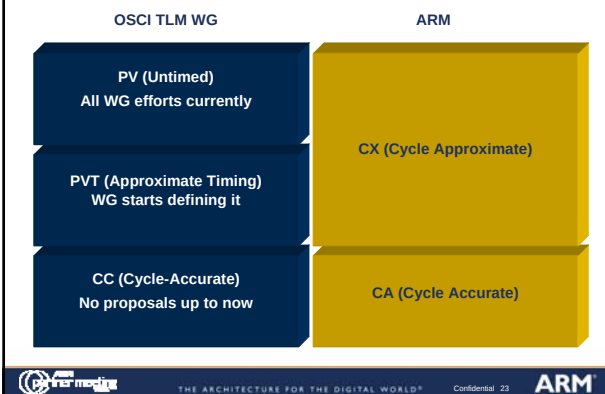
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## Abstraction Layers: OSCI TLM




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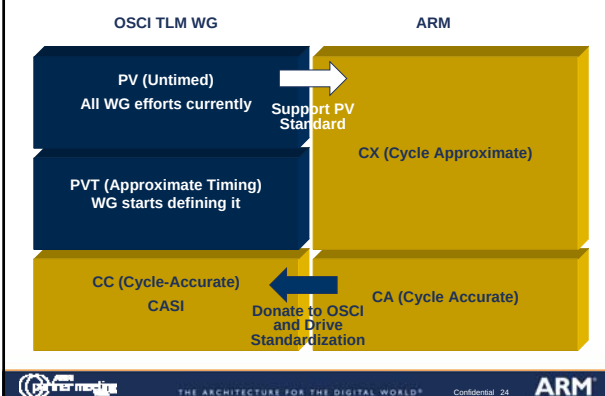
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## ARM is Adopting and Donating




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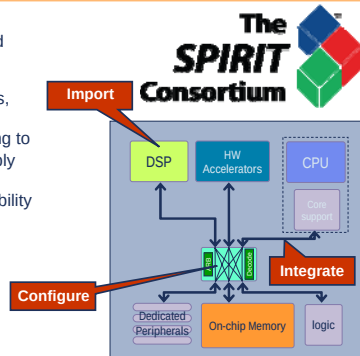
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## IP Packaging and Integration Standards

- Consortium for Standard Packaging, Integration and Reuse of IP in Tool flows
- Founded in 2003 by ARM, Cadence, Mentor Graphics, Philips, ST and Synopsys
- A standard for IP packaging to
  - Accelerate SoC assembly and verification
  - Promote tool interoperability
- Meta-data descriptions to enable...
  - IP import
  - IP configuration
  - IP integration
- ...at multiple levels of abstraction, ESL and RTL



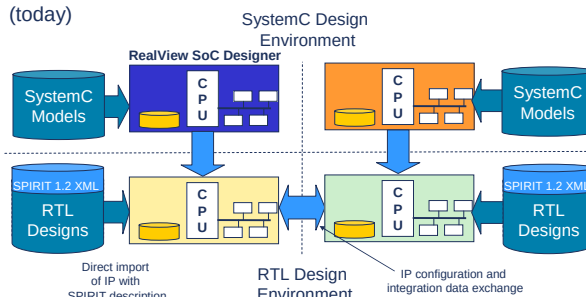
## The Growing Consortium Membership



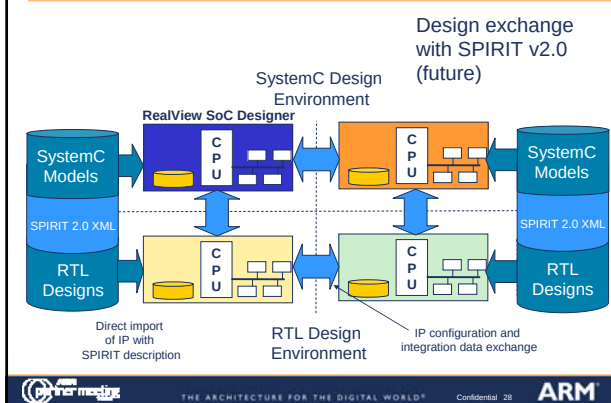
and new members: Mercury Computer Systems, NewLogic Technologies  
Soon including more than 50 members!

## SPIRIT for System Import/Export

Design exchange with SPIRIT v1.2 (today)



## SPIRIT for Full ESL Design Exchange



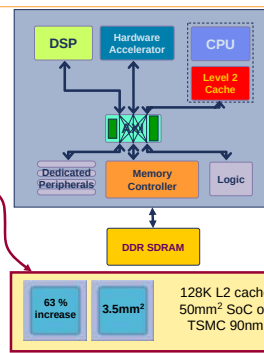
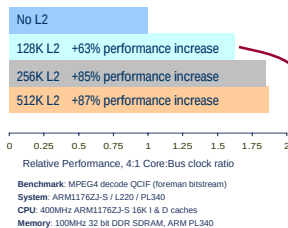
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## Typical Design Challenge

What cache size do I need for MPEG4 decode?

- Requires significant number of simulation cycles
- Hours becomes minutes in ESL



Best performance / cost trade-off

## ESL Tackles the Design Challenge

A feature-rich environment for virtual system prototyping helps...

- Validate system architectures against product requirements prior to RTL implementation
- Reduce cost by avoiding over-designing
- Eliminate silicon re-spins by avoiding under-design
- Parallelize hardware and software design tasks, accelerating time to market
- Exploit the value of designing with third-party IP



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## Samsung Proof Point

The screenshot shows the Samsung Semiconductor Global website. The main headline is "Samsung's VIP Design Methodology Reduces SoC Design Time Up to 40 Percent". The article text includes: "SEOUL, Korea - September 16, 2004 - Samsung Electronics Co., Ltd., a leader in advanced semiconductor technology, today reveals a hardware/software co-design methodology that will reduce overall design time for the company's System-on-Chip (SoC) products by up to 40 percent. The Virtual IP (VIP) approach will ultimately result in faster production of Samsung logic components, thus providing customers a time-to-market advantage." The article also mentions that Samsung developed the VIP design methodology in partnership with EDA vendors to address the ever-increasing complexity of SoC design challenges. It states that today's exploding market for multi-function consumer products requires more on-chip firmware as well as transistors, complicating and extending the design process. Using the VIP methodology, simulation speeds are hundreds or even thousands of times faster than that of a traditional RTL approach while maintaining more than 90 percent of accuracy over RTL. The article concludes that with the conventional RTL design methodology, it was not possible to design and optimize software before the field programmable gate array (FPGA) or prototype is implemented, which led to costly errors and delays. By using the VIP approach, software designers can now precisely fully-optimize and error-checked software before the development of RTL, reducing the SoC design time by up to 40 percent.

Thank you



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